Ludwing Valdés Reyes

Software Engineer · Backend & Enterprise Systems
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Portfolio: https://claude.ai/artifact/HUhWRnU9yRW2mFfuPD88b3 | GitHub: https://github.com/lilpne25-ui

PROFESSIONAL SUMMARY

Software engineer focused on backend, integrations and business systems. I turn manual processes into software that plant and office teams use every day, connected to the ERP, CAD and databases a company already runs. My CAD-to-ERP integration cut the time to load an assembly from 9 minutes to under 1.5.

WORK EXPERIENCE

Co-founder · Technology & product (personal project)
VLUX | Aug 2026 – Present

- Lead technology for four business product lines: CRM, point of sale, industrial costing and quality management.
- Design and build integrations with ERPs, CAD/PDM, legacy databases, messaging webhooks and devices.

IT & Systems
Innovax | Sep 2024 – Aug 2026

- Automated BOM loading from CAD to the ERP: 9:08 a 1:28 min per assembly on 6 workstations, every write verified.
- Built the attendance and overtime system that takes biometric clocks to SQL Server and delivers HR reports in their own template.
- Owned support, networks, servers, backups and equipment maintenance in an ISO 9001 operation.

TECHNICAL SKILLS

Languages: Python, C# / .NET 8, TypeScript, SQL
Applied AI: AI to speed up real systems: automating flows, analyzing data and accelerating development
Backend & cloud: Node.js, REST APIs, webhooks, Supabase, Cloudflare, Vercel deployments
Data: SQL Server, PostgreSQL, Pervasive PSQL, Prisma; Excel data cleanup and material standardization
ERP & CAD: Global Shop, Odoo Community (custom add-ons); SolidWorks and AutoCAD APIs to generate BOMs
Frontend & desktop: React, Next.js, Electron, WinForms, PySide6 / Qt, PWA
Infra & quality: Networks, servers, backups, Git; pytest, Vitest, Playwright; ISO 9001 support

SELECTED PROJECTS

CAD-to-ERP integration
In production | Technologies: Python, C# / .NET 8, SolidWorks API, SQL Server

- **Goal:** Remove the manual, error-prone BOM entry between engineering and the ERP.
- **My part:** Designed and built the whole app, from reading SolidWorks/PDM to validated writes in the ERP and change emails.
- **Result:** About 6× faster end to end (9:08 a 1:28), backed by 1,737 automated tests.

Commercial CRM
In operation | Technologies: Next.js, TypeScript, Prisma, PostgreSQL

- **Goal:** One traceable flow from lead to quote, project and warranty for a B2B installer.
- **My part:** Architecture and full-stack build: guided quoting, price catalog, idempotent web intake.
- **Result:** Web leads arrive on their own without duplicates, and no quote leaves without human review.

Attendance & overtime
In operation | Technologies: C#, .NET 8, SQL Server, ZKTeco SDK

- **Goal:** Stop HR from rebuilding clock data by hand every period.
- **My part:** Layered .NET 8 solution: clock sync to SQL Server and an overtime rules engine.
- **Result:** HR gets period reports in Excel and PDF, in their own template.

Also: ISO 9001 quality management system (demo), real-estate operations prototype and a point of sale on Odoo (pilot candidate). Full breakdown in the portfolio.

PORTFOLIO

Case studies with screenshots, metrics and technical detail: https://claude.ai/artifact/HUhWRnU9yRW2mFfuPD88b3